

Sleep Stage Classification from EEG Signals Using a Lightweight Convolutional Neural Network

Minor Project — AI for Signal Processing

1. Project Summary

This project develops a lightweight deep learning model that classifies sleep stages, Wake, N1, N2, N3, and REM, directly from raw EEG signals recorded during sleep. The system is trained and evaluated on the Sleep-EDF dataset, a free, expert-annotated polysomnography (PSG) dataset available through PhysioNet, which includes EEG channels alongside EOG and submental EMG signals, so no data collection or hardware is required.

The signal pipeline segments the EEG recording into standard 30-second epochs (the clinical scoring unit used by sleep experts), normalizes each epoch, and classifies it into one of the five sleep stages using a compact CNN. The project is entirely software-based and reproducible on a laptop. A night-long streaming simulation over static recordings, plotting the predicted sleep stage epoch by epoch, demonstrates how the model would behave in an overnight monitoring context.

2. Novelty

The core method, CNN-based classification of EEG epochs into sleep stages, is well established. The novelty of this project lies in the following additions:

- Single-channel, lightweight design: rather than the six-channel, image-based approach used in the base paper, this project targets a single EEG channel with a compact 1D-CNN, aligning with the practical direction of low-power, wearable-friendly sleep monitoring devices rather than full clinical polysomnography setups.
- Subject-independent evaluation: the dataset is split by subject rather than by epoch, so the model is tested on entirely unseen individuals, giving an honest measure of generalization instead of inflated within-subject accuracy.
- Stage-transition-aware error analysis: rather than reporting a single accuracy figure, confusion is analyzed specifically around adjacent, easily confused stages (particularly N1, which is well documented in the literature as the hardest stage to classify), giving a more clinically meaningful evaluation.
- Overnight monitoring simulation: predicted stages are plotted continuously across a full night's recording as a hypnogram, replicating the visual format sleep clinicians actually use, rather than only reporting a static test accuracy.

3. Real-World Relevance

Automated sleep staging has direct, well-documented applications:

- Sleep disorder diagnosis: conditions such as insomnia, sleep apnea, and narcolepsy are diagnosed through overnight PSG scoring, which is currently done manually by trained technicians and is time-consuming; automated staging reduces this clinical workload.

- Consumer wearable sleep tracking: single-channel, low-power EEG models are directly relevant to the growing market of wearable sleep-tracking headbands and devices, which need lightweight models that can run on limited hardware rather than full clinical CNN pipelines.
- Mental health and cognitive research: sleep architecture (time spent in each stage) is linked to mood disorders, memory consolidation, and stress; researchers use automated staging to process large volumes of overnight data that would be impractical to score manually.
- Remote and home-based patient monitoring: reduces the need for costly in-lab sleep studies, especially relevant for elderly care and rural or low-resource healthcare settings.

4. Sustainable Development Goal (SDG) Alignment

SDG 3	Good Health and Well-Being — automated, low-cost sleep monitoring directly supports early detection of sleep disorders and broader mental and physical health outcomes linked to sleep quality.
SDG 9	Industry, Innovation and Infrastructure — lightweight, single-channel model design reflects the broader push toward efficient AI deployable on low-power wearable health devices rather than cloud-dependent clinical systems.

5. Objectives

- Preprocess and segment raw EEG signals from the Sleep-EDF dataset into standard 30-second epochs.
- Design and train a lightweight single-channel 1D-CNN to classify epochs into the five AASM sleep stages (Wake, N1, N2, N3, REM).
- Evaluate the model using a subject-independent train/test split, reporting per-class precision, recall, F1-score, and confusion matrix, with particular attention to N1/REM confusion.
- Build a hypnogram-style visual simulation displaying predicted sleep stages across a full night's recording.

6. Dataset

Sleep-EDF Database (PhysioNet): polysomnography recordings including EEG (Fpz-Cz and Pz-Oz channels), horizontal EOG, and submental EMG, with expert-annotated sleep stage labels for each 30-second epoch, following AASM/Rechtschaffen and Kales scoring conventions. Publicly available without restriction, ensuring reproducibility.

7. Base Paper (Journal)

- [1] I. S. Masad, A. Alqudah, and S. Qazan, “Automatic Classification of Sleep Stages Using EEG Signals and Convolutional Neural Networks,” PLOS ONE, vol. 19, no. 1, e0297582, 2024. DOI: 10.1371/journal.pone.0297582.

This paper is the anchor reference for the project. It uses the Sleep-EDF dataset (via the same source repository) with the AASM five-stage scoring scheme, proposing a CNN-based methodology that converts EEG epochs into time-frequency images across six channels, achieving up to 99.39 percent accuracy on a single channel (C4-A1). The paper explicitly identifies stroke patient monitoring and general sleep quality studies as target applications, directly supporting the real-world relevance of this project. This project adapts the same classification task and dataset to a lighter single-channel, raw-signal 1D-CNN design more appropriate for a minor project scope and for wearable-oriented deployment.

8. Reference Papers (Journals)

- [2] C. Yang, B. Li, Y. Li, Y. He, and Y. Zhang, “LWSleepNet: A Lightweight Attention-Based Deep Learning Model for Sleep Staging with Single-Channel EEG,” *Digital Health*, vol. 9, 2023.
- [3] S. Zhang and X. Wang (Journal of Mechanics in Medicine and Biology), “EEG-Based Sleep Stage Classification for Entrepreneurial Students: A Lightweight and Efficient Deep Learning Approach,” *Journal of Mechanics in Medicine and Biology*, 2025.
- [4] H. Zhu et al., “A Deep Learning Method Approach for Sleep Stage Classification with EEG Spectrogram,” *International Journal of Environmental Research and Public Health*, vol. 19, no. 10, p. 6322, 2022. DOI: 10.3390/ijerph19106322.
- [5] Employing a Convolutional Neural Network to Classify Sleep Stages from EEG Signals Using Feature Reduction Techniques, *Algorithms*, vol. 17, no. 6, p. 229, 2024. DOI: 10.3390/a17060229.

9. Expected Outcome

A trained, evaluated lightweight single-channel EEG sleep stage classifier with documented per-class performance metrics, a confusion matrix highlighting N1/REM confusion patterns, and a working hypnogram-style overnight visual demo, packaged with a short technical report suitable for minor project submission and viva presentation.